Test report 15 – Sketches: corners, offset, stroke

Plan 17.7, step 2: `corner`, `fillet2d`/`chamfer2d`, `offset`, `stroke` · cad-kernel (`region2.rs`, biarcs in `curve2.rs`), cad-ir, cad-eval, cad-engine (`op2d.rs`), cad-session, cad-cli, web · 2026-10-23 · branch worktree-agent-a4870ff288873dafe · commit bf2257a · rustc 1.97.0 (2d8144b78 2026-07-07) · 4 cores

cargo test --workspace	clippy · fmt	Vitest · Playwright	OpenSCAD · STEP	offset per case	Everyday · Wasm (Node)
481 / 481 29 of them new/extended	**clean · clean** -D warnings	**215 · 97 / 98** new: E01–E05	**36/36 · 58/58** compare.py · verify.py (OCCT)	**≤ 0.026 %** 24 cases against OpenSCAD	**unchanged · ok** check.sh · test-node.mjs

1 Task and result

Step 2 of plan 17: **corners** in sketches (`corner(r = …)`, `corner(chamfer = …)`) and on arbitrary 2D profiles (`fillet2d`, `chamfer2d`, with zones as in 3D), **`offset`** compatible with OpenSCAD and exact for lines and arcs, **`stroke`** for open paths; display and handles in UI and CLI. The basis is a new module `cad_kernel::region2`: exact Booleans of regions made of lines and circular arcs, from which offset, stroke and the evaluation of arbitrary 2D subtrees are built. Text, SVG import (step 3) and draw mode (step 4) are not part of this.

- Areas against formulas (section 7): all 18 cases within the volume integration of SMLib (≤ $1.1 \cdot 10^{-7}$ relative) – rectangle with filleted corners = $ab - (4 - \pi)\, r^2$, offset of a square = $(a + 2r)^2 - (4 - \pi)\, r^2$, stroke of a circular arc = annular sector + caps.
- OpenSCAD 2021.01 (section 4): 24 individual cases, area deviation ≤ 0.026 % (polygonal, i.e. `delta` on polygons: ≤ 0.0002 %), bounding box ≤ 0.0009 mm; compare.py with the new models 34–36: 36/36. The remaining deviation for `r` is OpenSCAD's faceting (`$fn = 128`): we compute exact arcs.
- STEP: arcs as `CIRCLE`, extruded arcs as cylinders, no B-spline curves and no approximated edges in the new examples; OpenCASCADE reads them with the same volume (verify.py).

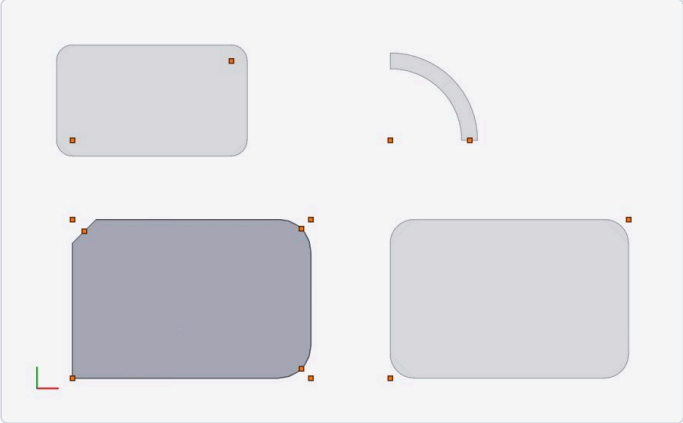
2 corner() in sketches

Rules (plan 17.9): `corner()` between two segments rounds the corner between them – the closing line of `close()` counts as a segment, so `corner()` before `close()` rounds the corner to the closing line; `corner()` **directly after `start(p)`** rounds the start corner (between the last and the first segment); likewise `corner()` after the last segment of a contour that ends at the start (no closing line needed). The sketch is first built without corners (tangential arcs and `line(length)` refer to the sharp corner), then all corners of a contour are computed on the original segments; if two corners need overlapping parts of one segment, both fail with the largest common value – independent of the order. The fillet is a segment of the corner line (highlighting); a start corner is placed at the end of the contour. Chamfer: intersection points at distance `d` from the corner point.

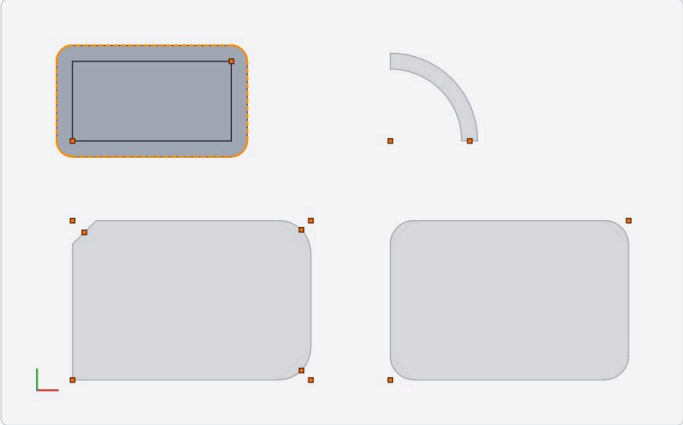
Segments	Solution
line – line, line – arc, arc – arc	exact: center = intersection of the curves offset by r (line/line, line/circle, circle/circle with R ± r depending on the side), foot points on the segments, shortest candidate
with Bézier or spline	Newton on $a(u) + r \cdot n_a(u) = b(w) + r \cdot n_b(w)$ (start value from the tangents), stop at residual 10^{-13} of the size, warning above 10^{-6}; the curve is trimmed at the parameter (stays a B-spline). Test: line–Bézier, residual < 10^{-9}, distance center–curve = r to 10^{-6}

Errors at the corner line: radius does not fit ("at most ≈ r", via bisection), tangential corner ("no corner"), cusp (180°), after `close()`, before `start`, twice at one corner, start/end of an open contour, value ≤ 0. **Handle:** radius at the middle of the fillet, base at the corner point, linear in r (UI test E01).

```
// Sketches step 2: corner, fillet2d, offset, stroke
linear_extrude(2) sketch {
    start([0, 0]);
    line(to = [30, 0]);
    corner(r = 4);
    line(to = [30, 20]);
    corner(r = 4);
    line(to = [0, 20]);
    corner(chamfer = 3);
    close();
}
translate([40, 0, 0]) linear_extrude(2) fillet2d(3)
square([30, 20]);
translate([0, 30, 0]) linear_extrude(2) offset(r = 2)
square([20, 10]);
translate([40, 30, 0]) linear_extrude(2) stroke(2,
cap = "flat") sketch(closed = false) {
    start([10, 0]);
    arc(center = [0, 0], angle = 90);
}
```



`cad render --line 5 --view top --show handles`: the sketch highlighted, handles (radius on the fillets)



`--line 13`: offset – source contour, offset contour dashed

```
linear_extrude(1) sketch {
    start([0, 0]);
    line(to = [20, 0]);
    corner(r = 12);
    line(to = [20, 8]);
    corner(r = 1);
    arc(to = [10, 18]);
    line(to = [0, 8]);
    corner(r = 2);
    corner(r = 3);
    close();
}
translate([30, 0, 0]) linear_extrude(1) fillet2d(1.5) square([10, 2]);
translate([50, 0, 0]) linear_extrude(1) offset(r = -6) square(10);
translate([70, 0, 0]) linear_extrude(1) offset(1) sketch(closed = false) { start([0, 0]); line(to = [5, 0]); }
```

```
cases/fehler.scad:10:5: Fehler (Auswertung): an dieser Ecke steht schon ein `corner()`
cases/fehler.scad:4:5: Fehler (Auswertung): corner(): Radius 12 passt nicht zwischen die Segmente (höchstens ≈
8)
cases/fehler.scad:6:5: Fehler (Auswertung): corner(): keine Ecke, die Segmente gehen tangential ineinander über
cases/fehler.scad:13:41: Fehler (Kernel): Ecke bei [0, 0]: Radius 1.5 passt nicht (höchstens ≈ 1)
cases/fehler.scad:13:41: Fehler (Kernel): Ecke bei [10, 0]: Radius 1.5 passt nicht (höchstens ≈ 1)
cases/fehler.scad:13:41: Fehler (Kernel): Ecke bei [10, 2]: Radius 1.5 passt nicht (höchstens ≈ 1)
cases/fehler.scad:13:41: Fehler (Kernel): Ecke bei [0, 2]: Radius 1.5 passt nicht (höchstens ≈ 1)
cases/fehler.scad:14:41: Warnung (Kernel): offset(-6) lässt vom Profil nichts übrig
cases/fehler.scad:15:23: Fehler (Kernel): offset: offene Skizze ist kein Profil (für Pfade `stroke(w)`
verwenden)
cases/fehler.scad:15:41: Fehler (Kernel): offset: offene Skizze ist kein Profil (für Pfade `stroke(w)`
verwenden)
cases/fehler.scad: 9 Fehler, 1 Warnungen
```

3 fillet2d and chamfer2d

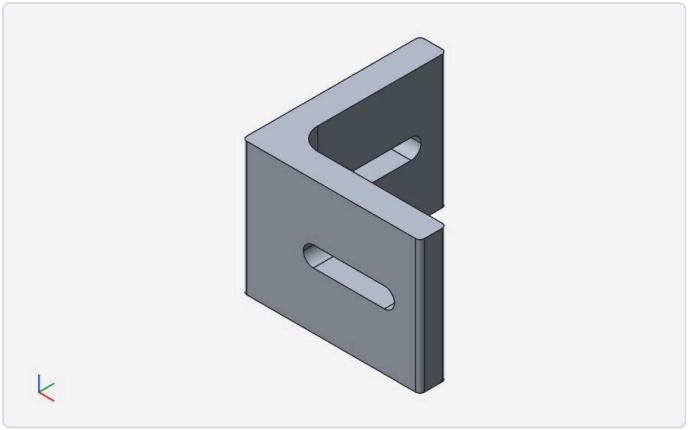
`fillet2d(r) { profil; zone; … }` / `chamfer2d(d)`: the first child is the profile (any origin: `square`, `polygon`, sketches, 2D Booleans), further children are zones (2D shapes and `zone(linie | ebene, d)`). Corners are joints with a kink, convex and concave; the fillet lies on the inner side of the kink (convex: material removed, concave: material added). **Selection** as in 3D: `select = "inside"` (default) – the corner point lies in a zone; `"touch"` – also when only the replaced piece of the profile touches a zone. Inline hint "n Ecken" ("n corners"), an empty selection is a warning, radii that are too large are errors per corner (the remaining corners are rounded). Nested blocks act from the inside out; the transitions of a fillet are tangential and do not count. A single closed contour keeps its segments (splines stay splines); everything else goes through the exact region.

```
// Purpose: angle bracket with a filleted profile
(plan 17.3) – L profile as a polygon; `fillet2d`
// with a zone rounds only the inner corner with a
large radius (notch effect), an outer `fillet2d`
// rounds all other corners with a small radius (the
transitions of the large arc are tangential
// and do not count); extruded, with slots from a
`stroke` of a line segment (round ends).
t = 6;       // wall thickness
a = 40;      // leg length
h = 30;      // width of the bracket
r_innen = 5;

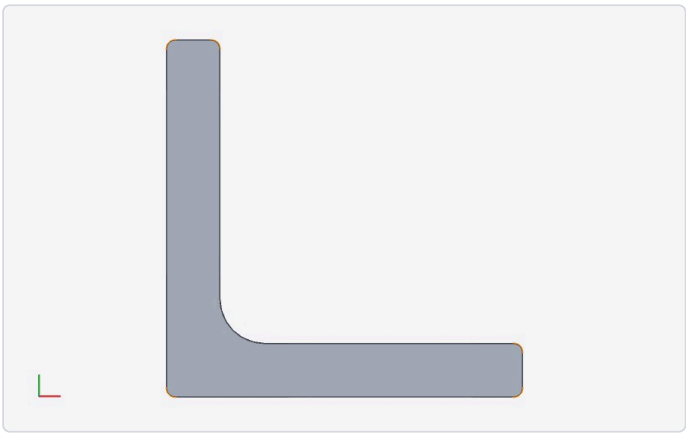
module profil() polygon([[0, 0], [a, 0], [a, t], [t,
t], [t, a], [0, a]]);

module langloch(von, bis, d = 4.5) stroke(d)
sketch(closed = false) {
    start(von);
    line(to = bis);
}

difference() {
    linear_extrude(h) fillet2d(1) {
        fillet2d(r_innen) {
            profil();
            translate([t, t]) circle(1);   // zone:
the inner corner
        }
    }
    // slots in the base (xz plane, through the wall
in y)
    translate([0, -1, 0]) rotate([-90, 0, 0])
linear_extrude(t + 2) langloch([16, -h / 2], [32, -h
/ 2]);
    // and in the wall (yz plane, through the wall in
x)
    translate([-1, 0, 0]) rotate([90, 0, 90])
linear_extrude(t + 2) langloch([16, h / 2], [32, h /
2]);
}
```



skizze_halter.scad: inner corner r 5 (zone), all other corners r 1, slots from `stroke`



`--line` 18 from above: the five new corner arcs of the outer block

```
Kontext: Datei
Z.18: linear_extrude(h) fillet2d(1) {
  [2] linear_extrude  Z.18:5–23:6  ok
      Hülle [0, 0, 0] – [40, 40, 30]
      Volumen 13448.7611 mm³, Fläche 5567.8233 mm²
  [3] fillet2d  Z.18:23–23:6  ok
      Hülle [0, 0, 0] – [40, 40, 0]
      ausgewählte Kanten: 5
      Hinweis: 5 Ecken
  Griff Height #0: Wert ["30"] bei [0, 0, 30] (1 Literal(e))
```

4 offset: semantics and comparison with OpenSCAD

As in OpenSCAD 2021.01: first positional argument `r`, without arguments `r = 1`, `r` wins over `delta`, `chamfer` only with `delta`. `r > 0`: convex corners become **exact arcs**, concave ones stay sharp; `r < 0`: convex ones stay sharp, concave ones become round, narrow parts disappear, holes grow. `delta`: corners extended to the intersection of the offset edges (without a miter limit – OpenSCAD sets `miter_limit` 10^6, as the acute triangle below confirms), `chamfer = true` cuts like Clipper's `jtSquare` perpendicular to the angle bisector at distance `delta` from the corner point (measured in OpenSCAD: 196 − 4·(2√2 − 2)² for the square). Exact: region ∪ (bands ± r along the segments ∪ circles at the corners), or region \ (…) for r < 0 – the Minkowski sum with the circle, made of lines and arcs; `delta`: one-sided bands and corner pieces (miter/chamfer) at diverging corners.

One file `linear_extrude(1)` … per case (volume = area, `reports/tmp/report15/openscad/`); OpenSCAD exports STL (`$fn = 128` for roundings, as in `make-reference.py`), we measure exactly (`cad measure`). "exact": closed-form formula where one exists.

Case	OpenSCAD mm²	ours mm²	dev.	exact	OpenSCAD − exact	bbox Δ mm
offset(r = 2) square(10)	192.5613	192.5664	0.0026 %	192.5664	-0.0026 %	0.0000
offset(r = −2) square(10)	36.0000	36.0000	0.0000 %	36.0000	0.0000 %	0.0000
offset(3) circle(r = 5) – positional = r	201.0091	201.0619	0.0263 %	201.0619	-0.0263 %	0.0009
offset(r = 1.5) L profile (concave corner stays sharp)	382.5822	382.5857	0.0009 %	–	–	0.0000
offset(r = −1.5) L profile (concave corner becomes round)	145.4836	145.4829	-0.0005 %	–	–	0.0000
offset() square(10) – r = 1	143.1403	143.1416	0.0009 %	143.1416	-0.0009 %	0.0000
offset(r = 2, delta = 1) – r wins	192.5613	192.5664	0.0026 %	192.5664	-0.0026 %	0.0000
offset(delta = 2) square(10)	196.0000	196.0000	0.0000 %	196.0000	0.0000 %	0.0000
offset(delta = 2, chamfer = true) square(10)	193.2548	193.2548	0.0000 %	193.2548	-0.0000 %	0.0000
offset(delta = −2) square(10)	36.0000	36.0000	0.0000 %	36.0000	0.0000 %	0.0000
offset(delta = 1) star (10 points)	220.8551	220.8553	0.0001 %	–	–	0.0000
offset(delta = 1, chamfer = true) star	216.9401	216.9405	0.0002 %	–	–	0.0000
offset(delta = 1) acute triangle (5.7°): miter without limit	48.2047	48.2047	0.0002 %	–	–	0.0000
offset(delta = 1.5) L profile	385.0000	385.0000	0.0000 %	–	–	0.0000
offset(delta = −1.5) L profile	145.0000	145.0000	0.0000 %	–	–	0.0000
offset(delta = −1.5, chamfer = true) L profile	145.3860	145.3860	-0.0000 %	–	–	0.0000
Plate 30 × 20 with circular and square hole (without offset)	485.7547	485.7345	-0.0042 %	–	–	0.0000
offset(r = 1) plate: holes smaller	638.8831	638.8673	-0.0025 %	–	–	0.0000
offset(r = −1) plate: holes larger	326.3437	326.3186	-0.0077 %	–	–	0.0000
offset(r = 4.5) plate: both holes closed	1,113.5916	1,113.6173	0.0023 %	–	–	0.0000
offset(delta = 1) plate	639.7427	639.7257	-0.0027 %	–	–	0.0000
offset(r = 1.5) of two squares 2 apart: they grow together	322.5843	322.5884	0.0013 %	–	–	0.0000
offset(r = −1): 1.5 mm web disappears, two parts	128.3124	128.3117	-0.0005 %	–	–	0.0000

offset(r = 2) offset(r = −2): opening – corners round, narrow parts gone	357.9701	357.9743	0.0012 %	–	–	0.0002

Where a formula exists, the deviation of our values from it is the integration accuracy of SMLib (section 7); that of OpenSCAD is the faceting (circular arcs as 128-gons: chords inside, slightly less area) or Clipper's integer coordinates (chamfer of the square: $3 \cdot 10^{-7}$).

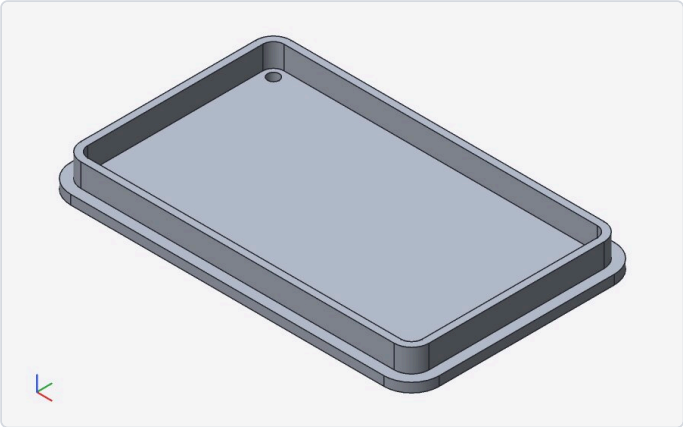
compare.py with the new models (`examples/openscad-compat/34_offset_round.scad`, `35_offset_delta.scad`, `36_offset_holes.scad`; reference generated with `make-reference.py` only for these files, `stl_stats.classify` counts `offset(r)` as rounded):

```
file                          class       V openscad       V ours    dV %    dA %  dBox mm  status
-------------------------------------------------------------------------------------------------
34_offset_round.scad          rounded       2989.182     2988.981   0.007   0.007   0.0009  ok
35_offset_delta.scad          polygonal     3076.875     3076.873   0.000   0.000   0.0000  ok
36_offset_holes.scad          rounded       7310.425     7310.656   0.003   0.001   0.0000  ok
```

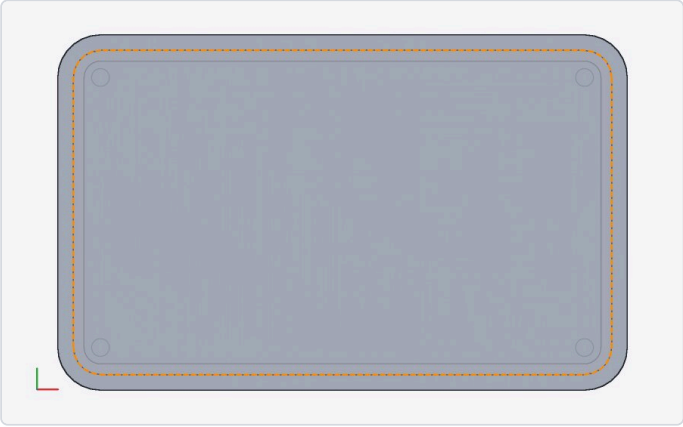
```
// Purpose: enclosure lid (plan 17.3) – the outline
is a sketch with rounded corners
// (`corner`), the plug-in rim a ring of two inward
offsets of the same outline
// (`offset(r = -…)`: the corner radii become smaller
but stay exact arcs), plus four
// screw holes in the corners.
l = 80;       // outer dimension
b = 50;
r = 6;        // outer corner radius
wand = 2;     // wall thickness of the enclosure
spiel = 0.2;  // clearance between plug-in rim and
wall
steg = 1.5;   // width of the plug-in rim

module umriss() sketch {
    start([0, 0]);
    corner(r = r);
    line(to = [l, 0]);
    corner(r = r);
    line(to = [l, b]);
    corner(r = r);
    line(to = [0, b]);
    corner(r = r);
    close();
}

difference() {
    linear_extrude(2) umriss();
    for (p = [[r, r], [l - r, r], [l - r, b - r], [r,
b - r]])
        translate([p[0], p[1], -1]) cylinder(h = 4, d
= 2.5);
}
translate([0, 0, 2]) linear_extrude(5) difference() {
    offset(r = -(wand + spiel)) umriss();
    offset(r = -(wand + spiel + steg)) umriss();
}
```



skizze_deckel.scad: outline with four `corner(r = 6)`, plug-in rim as a ring of two inward offsets



`--line 30`: source contour and offset contour (dashed); the corner radii 6 → 3.8 stay arcs

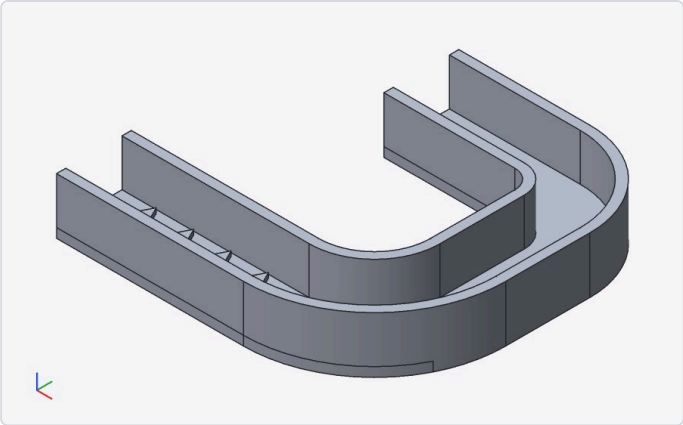
5 stroke

`stroke(w, cap = "round" | "flat" | "square", join = "round" | "miter" | "bevel")` (default round/round): children are open sketches (paths; a contour that ends at its start is closed) and arbitrary 2D profiles (their boundary as closed paths: `stroke(1) circle(5)` is a ring). Bands of half the width along the segments (line: rectangle, arc: annular sector), joins round, mitered (up to 4 half-widths like SVG's `stroke-miterlimit`, otherwise bevel) or beveled, ends round, flat or extended by w/2; crossing paths are united.

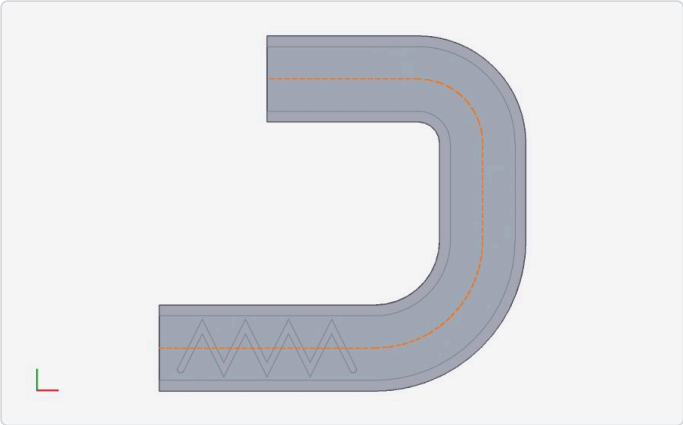
```
// Purpose: cable duct along a path (plan 17.3) – the
path is an open sketch of
// lines, a tangential arc and a rounded corner
(`corner`); `stroke` makes
// the floor from it (full width, flat ends) and the
walls (difference with a narrower
// stroke whose square ends reach beyond the duct:
the ends stay open); plus
// ribs in the floor as a zigzag with miter joins.
breite = 16;   // outside
wand = 2;
hoehe = 12;
boden = 2;

module pfad() {
    sketch(closed = false) {
        start([0, 0]);
        line(to = [40, 0]);
        arc(to = [60, 20]);          // tangential:
quarter circle r = 20
        line(to = [60, 50]);
        corner(r = 12);
        line(to = [20, 50]);
    }
}

linear_extrude(boden) stroke(breite, cap = "flat")
pfad();
linear_extrude(hoehe) difference() {
    stroke(breite, cap = "flat") pfad();
    stroke(breite - 2 * wand, cap = "square") pfad();
}
// ribs on the floor of the straight initial section
translate([0, 0, boden]) linear_extrude(1)
stroke(1.2, cap = "round", join = "miter")
sketch(closed = false) {
    start([4, -4]);
    for (i = [1 : 8]) line(to = [4 + 4 * i, i % 2 ==
0 ? -4 : 4]);
}
```



skizze_kabelkanal.scad: floor and walls from `stroke` of a path with a tangential arc and `corner(r = 12)`, ribs as a zigzag with miter joins



`--line 22`: center line of the floor

6 Kernel: exact 2D regions

- **`region2::combine(operanden, prädikat)`**: all segments are split at their exact intersection points (line–line, line–circle, circle–circle including tangency, overlap of collinear lines and identical circles; points within 10^{-9} of the size are merged), duplicate pieces are dropped. Each piece is kept where the predicate differs on its left and right (test points next to the middle of the piece; winding number per fill rule via the ray towards +x, arcs split into y-monotone pieces), with the region on the left. Joined into loops with the tightest turn at shared vertices; pieces of the same line or circle are merged. Result: outer loops counterclockwise, holes clockwise, only lines and arcs.
- 2D subtrees: the predicate is the Boolean expression of the leaves (fill rule per leaf: polygons even-odd, otherwise nonzero); a randomized test compares 60 Booleans of circles, rectangles, polygons and sketches with the polygon clipping of the display (≤ 10^{-3}).
- **B-splines** enter as G1 biarcs (two tangential arcs per piece, pieces halved until the deviation is ≤ 1 µm, at most 10^{-5} of the model size). Rationale: the offset of a B-spline is not a B-spline, the offset of an arc is an arc – the offset is exact with respect to the approximation and tangent-continuous, intersections stay solvable in closed form, STEP contains lines and circles; 1 µm is far below printing and manufacturing tolerances and above the computational accuracy.
- Engine: `Op::Offset`, `Op::Blend2`, `Op::Stroke` are 2D nodes; `profile::lift` computes the region (cached by subtree content) and passes it on as sketch leaves – extrude, revolve, loft, sweep and STEP see exact lines and arcs.

`cad_engine::region_2d` returns the region of a 2D node (bounding boxes for the compatibility tests).

Fix from step 1 (arcs over 90°). The facade builds an arc as a rational quadratic B-spline with spans ≤ 90°; a longer arc is a curve of several spans. Booleans with extrusions of such curves failed in SMLib depending on position (ring of two 180° arcs ∪ box, rotated by 157.5°: "volume 924.12 impossible for operands 976.79, 43.2"; correct with 4 × 90°). The engine now splits arcs over 90° into equal pieces ≤ 90° before the kernel (`profile::split_arcs`; extrude, revolve, sweep – not loft); a half circle becomes two cylindrical faces. Regression test `ring_of_long_arcs_unites_with_a_box`: arcs of 180°, 120°, 90° and 360°, each with eight box positions, volume = 3 × exact 2D area of the union (≤ 10^{-5}). Found along the way: the winding number of the region took arc end points from the angles (rounding ~10^{-15}), so a ray exactly through a vertex counted wrongly (ring of three 120° arcs without a hole) – now the shared vertices of the loop are used (test `rings_of_arcs_of_any_sweep`).

Model	Faces	CIRCLE	LINE	B-spline curves	Surface types	approximated
skizze_deckel	32	32	52	0	cylinder 16, plane 16	0 / 0
skizze_halter	26	28	44	0	cylinder 14, plane 12	0 / 0
skizze_kabelkanal	63	36	144	0	cylinder 16, plane 47	0 / 0
34_offset_round	57	40	95	0	cylinder 20, plane 37	0 / 0
36_offset_holes	105	74	193	0	cylinder 37, plane 68	0 / 0

OpenCASCADE (`tests/step/verify.py`) for the new models (columns: bodies, faces, V kernel, V OCCT, dV, dA, dBox, surface types):

```
model                      ok   solids faces       V kernel         V OCCT    dV rel    dA rel      dBox
surfaces (OCCT)
34_offset_round            ok        6    57        2989.32        2989.32 6.373e-09 8.755e-07 2.817e-16
{'cylinder': 20, 'plane': 37}
35_offset_delta            ok        8    76        3076.87        3076.87 2.956e-16 2.214e-16 0.000e+00
{'plane': 76}
36_offset_holes            ok        8   105        7310.33        7310.33 3.295e-08 3.248e-07 5.063e-16
{'cylinder': 37, 'plane': 68}
skizze_deckel              ok        1    32        9632.65        9632.65 3.306e-08 3.629e-06 1.502e-16
{'cylinder': 16, 'plane': 16}
skizze_halter              ok        1    26        12393.9        12393.9 3.224e-09 2.249e-08 2.219e-16
{'cylinder': 14, 'plane': 12}
skizze_kabelkanal          ok        1    63        9898.65        9898.65 5.552e-08 6.531e-07 8.018e-06
{'cylinder': 16, 'plane': 47}
```

7 Measurements: areas against formulas, run time

`linear_extrude(1)` … per case (`reports/tmp/report15/cases/`), `cad measure --volume` with the release build (time: whole CLI call without daemon). The geometry is exact; the deviation is that of the volume integration in SMLib.

Case	Kernel mm²	Formula mm²	rel. dev.	ms
`fillet2d(4) square([30, 20])`: ab − (4 − π) r²	586.2655	586.2655	$6.3{\cdot}10^{-9}$	51
same as a sketch with four `corner(r = 4)` (start corner directly after `start`)	586.2655	586.2655	$6.3{\cdot}10^{-9}$	57
`fillet2d(5) square(10)`: the lines vanish, a circle	78.5398	78.5398	$2.4{\cdot}10^{-8}$	58
`chamfer2d(2) square([30, 20])`: ab − 4 · d²/2	592.0000	592.0000	0	52
L profile, all corners r 2: 5 convex (−), 1 concave (+), each (1 − π/4) r²	252.5664	252.5664	$5.9{\cdot}10^{-9}$	56
same, only the concave corner (zone)	256.8584	256.8584	$2.1{\cdot}10^{-10}$	42
`offset(r = 2) square(10)`: (a + 2r)² − (4 − π) r²	192.5664	192.5664	$3.8{\cdot}10^{-9}$	55
`offset(r = 0.5) square(20)`	440.7854	440.7854	$8.0{\cdot}10^{-10}$	67

`offset(r = 1)` of a ring 10/4: π (11² − 3²)	351.8584	351.8584	$1.0 \cdot 10^{-8}$	179
`offset(delta = 2, chamfer = true) square(10)`: cut at distance 2	193.2548	193.2548	0	40
`stroke(2, cap = "flat")` of a quarter circle r 10: annular sector	31.4159	31.4159	$9.5 \cdot 10^{-8}$	43
same with round caps: annular sector + π (w/2)²	34.5575	34.5575	$1.1 \cdot 10^{-7}$	64
same with square caps: annular sector + w²	35.4159	35.4159	$1.1 \cdot 10^{-7}$	55
Polyline with a right angle, w 2, miter: bands + 1	40.0000	40.0000	0	40
same, round join: bands + π/4	39.7854	39.7854	$8.8 \cdot 10^{-10}$	55
same, bevel: bands + 1/2	39.5000	39.5000	0	41
`stroke(2) circle(10)`: ring π (11² − 9²)	125.6637	125.6637	$3.3 \cdot 10^{-9}$	143
Quarter circle r 10, `corner(r = 1)` between line and arc (closed form)	78.2860	78.2860	$3.3 \cdot 10^{-8}$	54

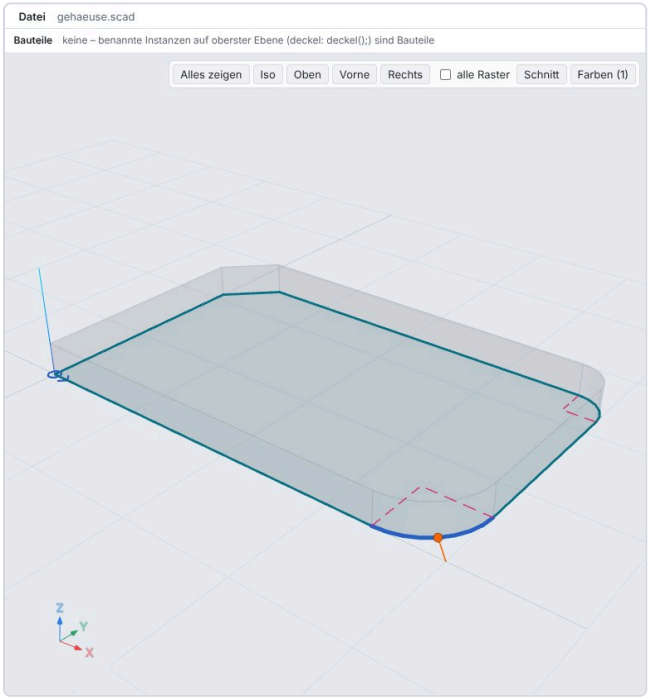
Run time (release, `cad check`, best of three runs, whole call): gear with 160 corners and a bore – `offset(r = 0.5)`, `fillet2d(0.4)` of all 160 corners – and `stroke` of a spline, 2D only: **74 ms**; the same extruded (kernel): 1,830 ms. Examples: lid 417 ms, bracket 213 ms, cable duct 935 ms (mostly kernel Booleans).

8 UI: display and handles

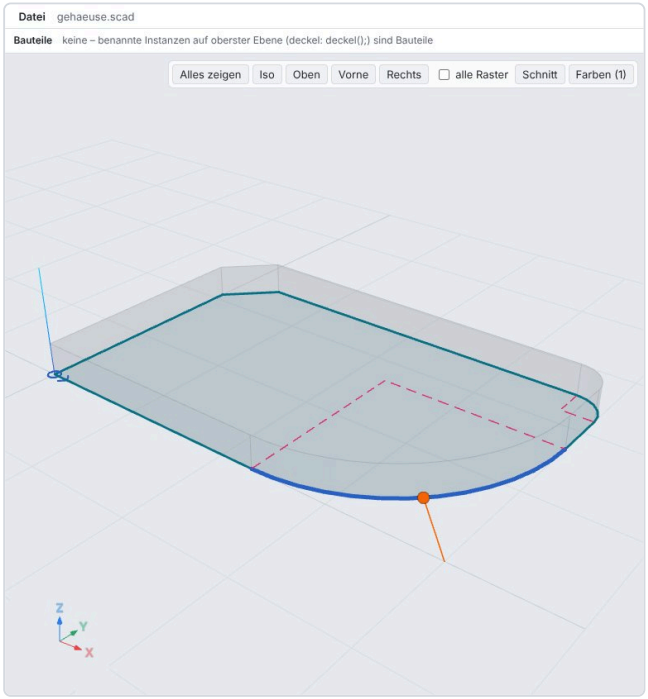
Cursor on `offset`: the children as an outline (source contour), the offset contour dashed in the accent color; on `stroke`: the path as an outline, the center line dashed; on `fillet2d`/`chamfer2d`: zones blue, the profile before filleting as an outline, the new corner pieces like selected edges, "n Ecken" ("n corners") at the line; on `corner`: the fillet as the active segment of the sketch with a radius handle. The helper lines come from the engine (`SceneNode.guides`: `source`, `result`, `centerline`). Checks via the UI state (`window.__cad`) and pixels of the layers (`guide:offset`, `guide:centerline`, `sketch:active`).

E01 Radius handle of corner()

Cursor on a corner line: the fillet is the segment of this line (thick in the accent color), the radius handle sits at the middle of the arc and moves linearly with r away from the corner point. Dragging (Alt: free) writes the literal; the neighboring segments are trimmed anew.



Cursor on corner(r = 6)



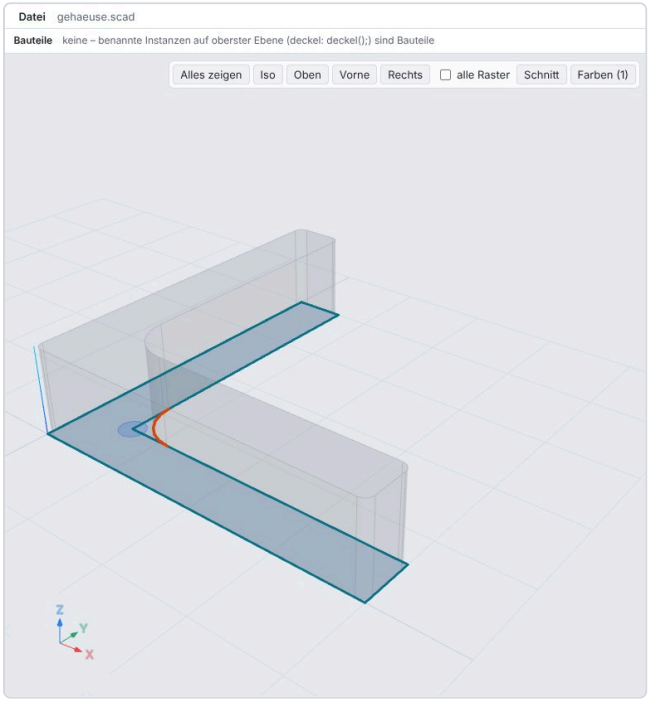
after dragging: corner(r = 17)

```
linear_extrude(3) sketch {
    start([0, 0]);
    line(to = [40, 0]);
    corner(r = 6);
    line(to = [40, 25]);
    corner(r = 3);
    line(to = [0, 25]);
    corner(chamfer = 4);
    close();
}
```

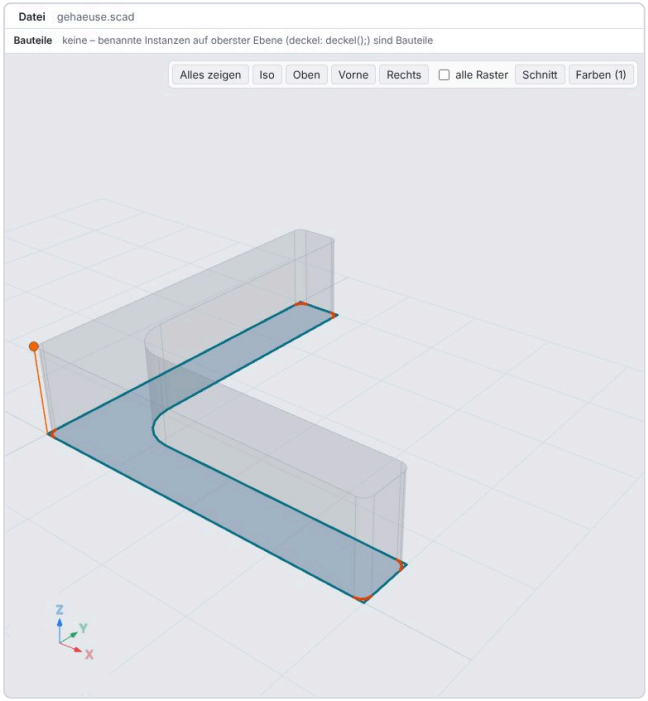
- handle: radius, value 6
- literal: 6 → 17
- pixels of the active arc: 835
- other lines unchanged

E02 fillet2d with a zone

The inner block rounds only the corner in the zone (circle around [t, t], blue) with r = 5; the outer one all other corners with r = 1 – the transitions of the large arc are tangential and do not count. The new corner arcs are colored like selected edges; the count appears as an inline hint at the line.



Cursor on fillet2d(5): zone around the inner corner



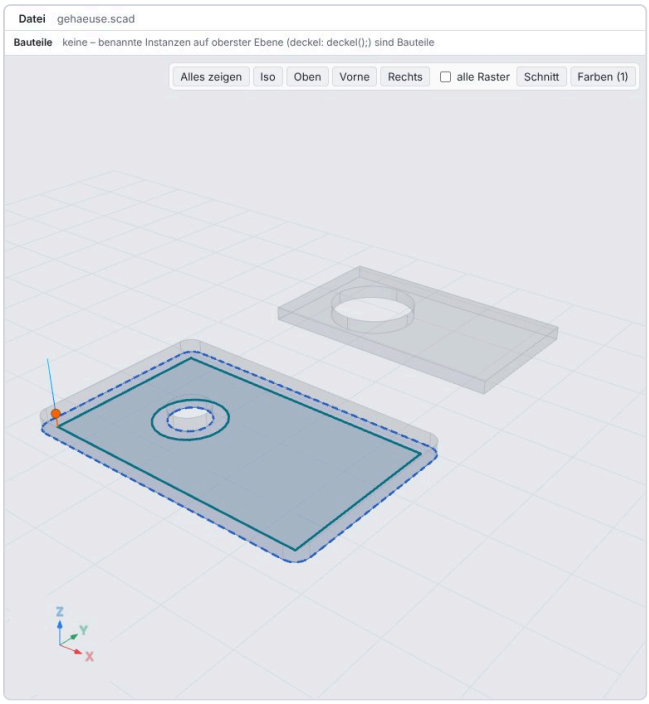
Cursor on fillet2d(1): the other five corners

```
t = 6;
linear_extrude(10) fillet2d(1) {
    fillet2d(5) {
        polygon([[0, 0], [40, 0], [40, t], [t, t], [t, 40], [0, 40]]);
        translate([t, t]) circle(1.5);
    }
}
```

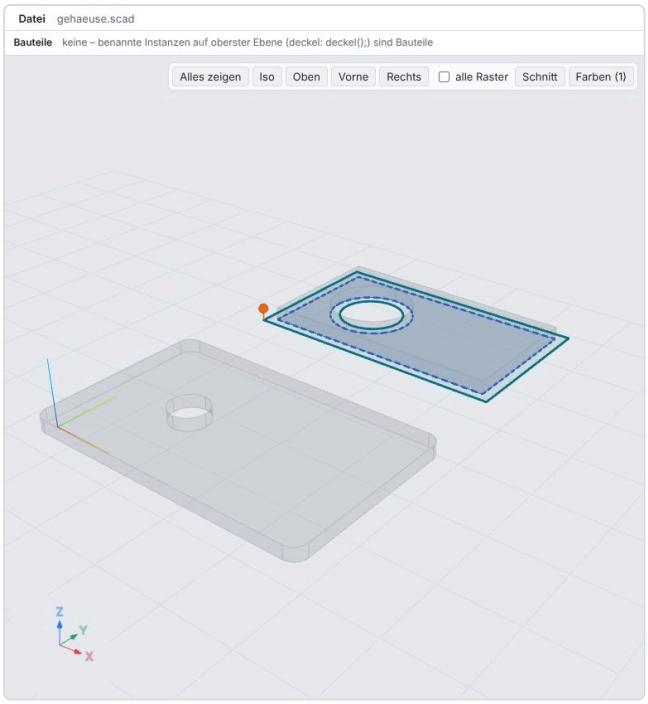
- zone: 1 (blue)
- hint line 3: 1 corner
- hint line 2: 5 corners
- selected edges (outer): 5

E03 offset: display

Cursor on offset: the profile of the children as an outline (source contour), above it the offset contour dashed in the accent color; the result itself as a face. Outward, the corners become exact arcs and the hole gets smaller; inward (delta), the corners stay sharp and the hole grows.



offset(r = 2): corners as arcs, hole smaller



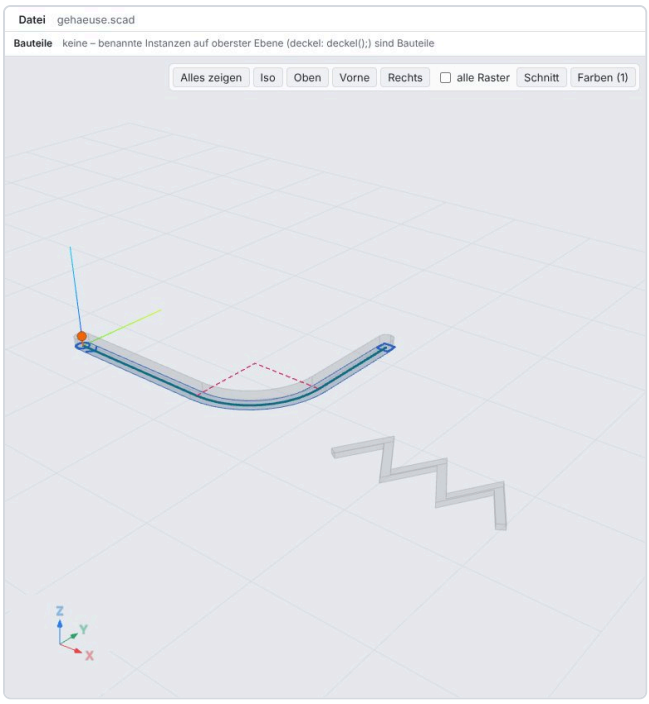
offset(delta = -1.5): hole larger, corners sharp

```
module platte() difference() {
    square([40, 24]);
    translate([12, 12]) circle(5);
}
linear_extrude(2) offset(r = 2) platte();
translate([0, 40, 0]) linear_extrude(2) offset(delta = -1.5) platte();
```

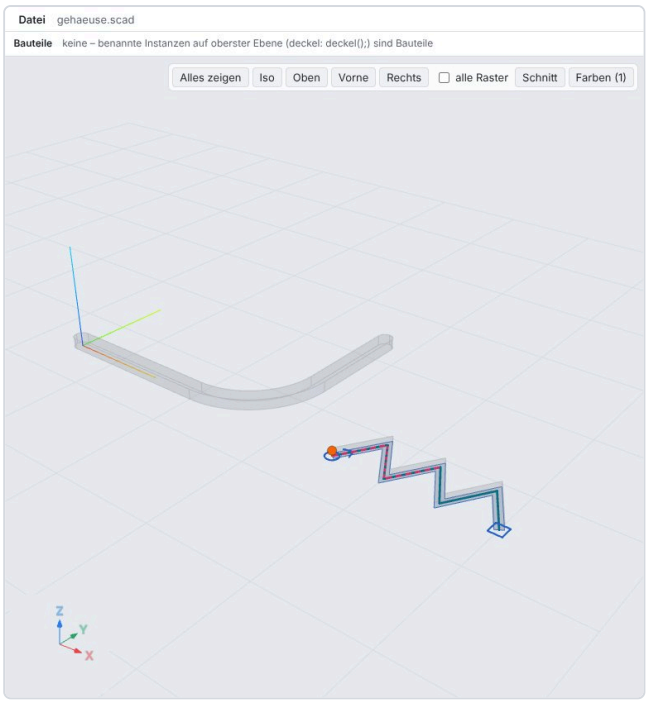
- offset contours: 2
- pixels: 2523
- bounding box −2 … 42 × −2 … 26
- inward: 2 contours

E04 stroke: display

Cursor on stroke: the path (open sketch) as an outline, the center line dashed, the result as a face. The arc becomes an exact annular sector, the caps half circles; the zigzag gets miters at the kinks and flat ends.



stroke(3) of a path of line, arc, line



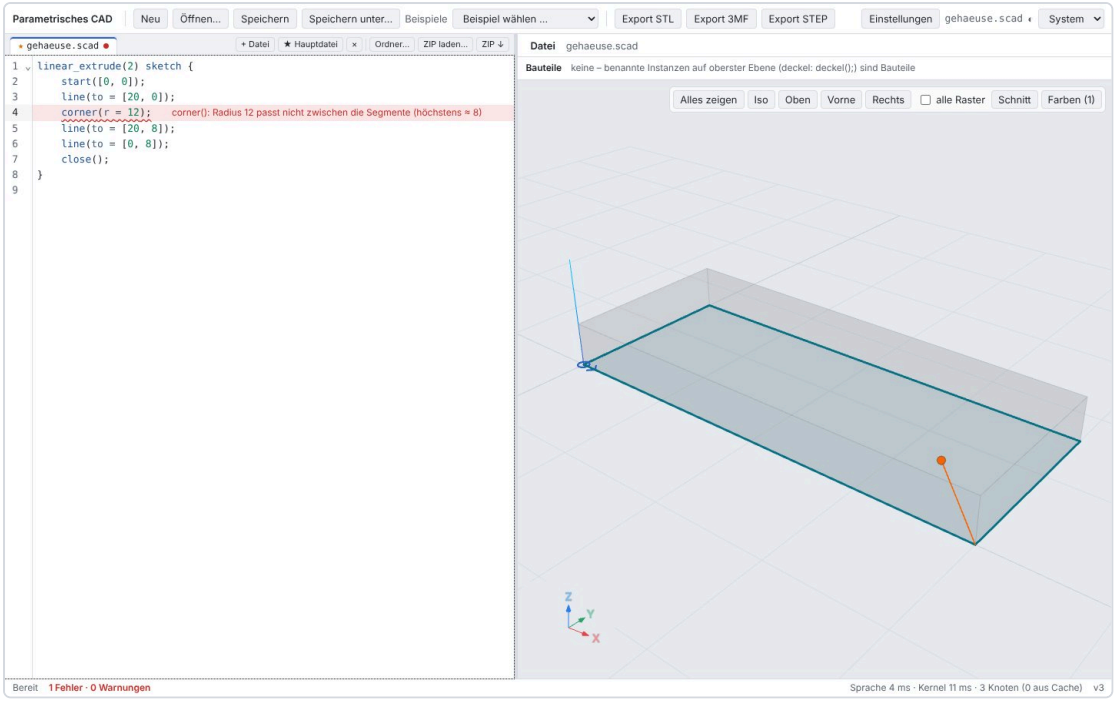
stroke(1.5, flat, miter) of a zigzag

```
linear_extrude(2) stroke(3, cap = "round") sketch(closed = false) {
    start([0, 0]);
    line(to = [30, 0]);
    arc(to = [45, 15]);
    line(to = [45, 35]);
}
translate([60, 0, 0]) linear_extrude(1) stroke(1.5, cap = "flat", join = "miter") sketch(closed = false) {
    start([0, 0]);
    for (i = [1 : 6]) line(to = [5 * i, i % 2 == 0 ? 0 : 8]);
}
```

- center lines: 1
- pixels: 184
- zigzag: 1 center line

E05 Radius does not fit

The radius does not fit on the short edge: error at the corner line with the largest fitting value; the sketch stays visible with the sharp corner, the handle sits at the corner point.



corner(r = 12) on an 8 mm long edge

```
linear_extrude(2) sketch {
    start([0, 0]);
    line(to = [20, 0]);
    corner(r = 12);
    line(to = [20, 8]);
    line(to = [0, 8]);
    close();
}
```

- 4: corner(): Radius 12 passt nicht zwischen die Segmente (höchstens ≈ 8) (radius 12 does not fit between the segments (at most ≈ 8))

9 Test runs

Run	Result
`cargo test --workspace --no-fail-fast`	481 passed, 0 failed
`cargo clippy --workspace --all-targets -- -D warnings` · `cargo fmt --check`	clean · clean
`pnpm typecheck` · `pnpm test` (Vitest)	clean · 215 passed
`pnpm e2e` (Playwright, Chromium)	97 / 98 – failed: C16 drag point handle constrained to an axis (Shift, X/Y/Z) · C16 (drag point handle constrained to an axis, report 3) ran into the test timeout (90 s) in the full run and passes repeatedly on its own (25 s): timing-dependent, not touched by this step.
`tests/openscad-compat/compare.py --eval`	36/36
`tests/step/verify.py` (OpenCASCADE)	58/58 passed
`tests/alltag/check.sh`	19 models, 0 with problems, 56 s, exports in /tmp/alltag-9jmnxaoh (deleted)
`node crates/kernel-smlib/js/test-node.mjs`	passed

New or extended Rust tests (29):

`sketch_corners_offset_stroke` **ok** · `op2d::tests::memo_key_depends_on_parameters` **ok** · `op2d::tests::exact_booleans_match_polygon_clipping` **ok** · `step_has_exact_lines_and_circles` **ok** · `ring_of_long_arcs_unites_with_a_box` **ok** · `concave_corners_and_zones` **ok** · `corner_radius_handle` **ok** · `arguments_of_2d_operations` **ok** · `corner_errors_at_the_line` **ok** · `corner_rules` **ok** · `curve2::tests::mapped_arcs` **ok** · `curve2::tests::interpolation_hits_points_and_tangents` **ok** · `region2::tests::fillet_bezier_numerically` **ok** · `curve2::tests::arc_nurbs_lies_on_circle` **ok** · `region2::tests::fillet_line_line` **ok** · `curve2::tests::join_keeps_the_geometry` **ok** · `region2::tests::biarcs_follow_the_curve` **ok** · `region2::tests::fillet_line_arc_and_arc_arc` **ok** · `region2::tests::offset_square_delta` **ok** · `region2::tests::rings_of_arcs_of_any_sweep` **ok** · `region2::tests::square_minus_circle_is_exact` **ok** · `region2::tests::stroke_of_a_polyline_joins` **ok** · `region2::tests::union_of_overlapping_squares` **ok** · `region2::tests::offset_square_round` **ok** · `region2::tests::offset_circle_and_hole` **ok** · `region2::tests::stroke_of_an_arc` **ok** · `openscad_models_stage1` **ok** · `spec_models_stage1` **ok** · `tree_bounds_match_openscad` **ok**

Vitest: `web/src/logic/sketch2d.test.ts` (offset contour, center line, zones and corner arcs of fillet2d, radius handle of corner on scenes of the examples). CLI: `sketch_corners_offset_stroke` (volume of all four operations, hint "4 Ecken", radius handle, helper lines in `render`, errors at the corner line). Compatibility: `openscad_compat`, `tree_bounds` (bounding boxes of the new models exactly via `region_2d`), `spec_models` (three new examples).

10 Limitations

- `offset(delta)` on arcs extends the tangents (miter of the tangent lines) – OpenSCAD only has polygons there. B-splines enter `offset`, `stroke` and 2D Booleans as biarcs (1 µm); the result contains arcs instead of the spline; `corner` and `fillet2d` of a single contour keep the spline.
- `fillet2d`/`chamfer2d` have no radius handle (only `corner` does); zones are 2D shapes and `zone(linie | ebene)`; 3D zones are ignored with a warning.
- The region classifies pieces via test points at a distance of ~10^{-6} of the size: boundaries that come closer than that without intersecting or overlapping can be classified wrongly. Candidate pairs via an x sort (quadratic in the worst case).
- 2D Booleans *between* results (`difference() { stroke(…); stroke(…); }`) remain 3D Booleans of the extruded parts as before; `offset`, `fillet2d`, `stroke` compute exactly in 2D (over their whole subtree).
- `stroke`: miter limit fixed at 4 half-widths; an arc with a radius below half the width is swept as a sector plus a mirrored sector. Errors of a 2D operation under an extrusion appear at both lines (as before for 2D errors).
- Lofts join sketches segment by segment and still receive arcs over 90° as curves of several spans; whether SMLib Booleans with such loft faces have the same weakness has not been checked.
- `cad render --line N` of a corner line highlights the whole sketch (highlighting of the single segment only in the UI, as in report 14).